

ME 336 Modeling and Control of Dynamic Systems

Spring-2025 Syllabus



Type of Course	Required	Prerequisites/Corequisites	ME 242
Credit Hours - ECTS	(3 0 0) 3 – 5 ECTS	Class Hours	Wed 15:00 – 18:00 (F206)
Instructor	Dr. Kutlu Kandemir	Instructor e-mail	kutlu.kandemir@tedu.edu.tr
Office	TBD	Office Hours	after lecture by appointment
Textbook & References	<u>Textbook:</u> · Nise, N.S., Control Systems Engineering, International Student Version, 6th Edition, John Wiley & Sons, New Jersey, 2011. ISBN: 9780470646120 <u>Reference books:</u> · Ogata, K., Modern Control Engineering, 5th International Edition, Prentice Hall, New Jersey, 2010. ISBN: 9780137133376 · Dorf, R.C., Bishop, R.H., Modern Control Systems, 10th Edition, Prentice Hall, New Jersey, 2005. ISBN: 0131277650 · Franklin, G.F., Powell, J.D., Emami-Naeini, A., Feedback Control of Dynamic Systems, 6th International Edition, Prentice Hall, New Jersey, 2008. ISBN: 9780135001509 · Palm, W.J., System Dynamics, 2nd International Edition, McGraw-Hill, 2010. ISBN: 9780071267793 · Ercan, Y., Mühendislik Sistemlerinin Modellenmesi ve Dinamiği, 2.Baskı, Literatür Yayınevi, İstanbul, 2003. ISBN: 9789750401077 · Efe, M.Ö., Otomatik Kontrol Sistemleri, Seçkin Yayınevi, 2012.		
Course Description	Modeling of physical systems and study of dynamic equations. Transfer functions and block diagrams. Fundamentals of automatic control. Control actions. Time response. Stability and Routh-Hurwitz criterion. Root-locus diagrams. Frequency response methods and Bode diagrams.		
Learning Outcomes	Having completed this course, students are expected to: 1. model dynamics of physical systems and express them by means of transfer functions and block diagrams. 2. know fundamental control architectures and apply basic feedback control actions (P, PD, PI, PID). 3. be able to perform stability and performance analysis of linear control systems. 4. draw and interpret the root-locus of a feedback system. 5. be able to determine the frequency response of a control system and use it to evaluate or adjust the stability and performance by means of Bode plots.		
Student Workload	Lectures 40 hrs Course readings 30 hrs Exams/Quizzes 50 hrs Other 20 hrs TOTAL 140 hrs		
Course Assignments	Midterm 35% Quizzes 25% Final 40% TOTAL 100% Quizzes will be conducted within the lecture hours. Number of quizzes will be at least 3. There will also be 5% bonus from attendance and participation in class		

Tentative Course Outline

Week	Topic
1	Introduction Modeling of physical systems and study of dynamic equations
2	Modeling of physical systems and study of dynamic equations
3	Modeling of physical systems and study of dynamic equations
4	Modeling of physical systems and study of dynamic equations Laplace transform
5	Transfer functions and block diagrams
6	Transfer functions and block diagrams Fundamentals of automatic control
7	Control actions - Case Study
8	MIDTERM
9	Time response
10	Time response Stability and Routh-Hurwitz criterion
11	Stability and Routh-Hurwitz criterion Root-locus diagrams
12	Root-locus diagrams
13	Frequency response methods and Bode diagrams
14	Frequency response methods and Bode diagrams

Course Policies & Remarks
I. Attendance You are expected to attend all classes. Classes start on the hour. Please be respectful of your classmates by being on time. Cell phones should be turned off and kept out of sight. Please do not use your computers during class time.
II. Calculator Policy You can use calculators but strictly NO mobile phones during exams.
III. Plagiarism For more information on TEDU policy on intellectual integrity see the link https://ds.tedu.edu.tr/sites/default/files/content_files/tedu_ogrenci-el-kitabi.pdf
IV. Disability Support If you have a disabling condition which may interfere with your ability to successfully complete this module, please contact Dr. Onur Özmen (e-mail: onur.ozmen@tedu.edu.tr). For more information see Handbook for Registered Students.
V. Make Up Exams A make-up exam for the Midterm will be provided by the instructor only if you have a legitimate reason for missing it. Acceptable reasons include illness or emergencies (e.g., a death in the family, a traffic accident, etc.). In such cases, you must provide a formal document to support your claim. The procedure for missing the Final exam is different. It is determined by the Make-up Exam Commission in accordance with the rules and regulations of TEDU. For more details see the following link: https://www.tedu.edu.tr/sites/default/files/docs/16-Nisan-2012-PAZARTESI_15082024.pdf